Multilayer Perceptron Based Classifier for FashionMNIST Dataset using PyTorch

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***Abstract*—This study investigates the use of a Multilayer Perceptron (MLP) neural network for classifying images in the FashionMNIST dataset using PyTorch. The network was trained using normalized image data with an architecture of fully connected layers and ReLU activations. The experiment evaluates performance using metrics like accuracy, precision, recall, F1-score, and AUC-ROC. The results indicate that a simple MLP can achieve high performance for image classification tasks. ROC curves and training-validation loss plots were also generated to visualize model performance and convergence behavior.**

***Index Terms*—Multilayer Perceptron, FashionMNIST, PyTorch, Classification, Deep Learning, ROC Curve**

I. Introduction

A. Literature Review

Image classification is a central task in computer vision, where the goal is to assign predefined labels to input images. The introduction of deep learning has significantly advanced this field, with neural networks achieving remarkable accuracy across a wide range of image datasets. One such benchmark dataset is FashionMNIST, introduced by Xiao et al. [1], as a more realistic alternative to the original MNIST digit dataset created by LeCun et al. [2]. FashionMNIST contains grayscale images of clothing items, offering increased complexity and diversity, thereby serving as a more effective benchmark for evaluating classification models.

Multilayer Perceptrons (MLPs), though simpler than Convolutional Neural Networks (CNNs), have proven to be effective in handling low-resolution image data [3]. They offer a foundational understanding of neural network training dynamics and serve as a baseline for more advanced architectures. The theoretical principles behind deep learning, including optimization algorithms like Adam and activation functions such as ReLU, are extensively detailed by Goodfellow et al. [4].

This experiment aims to develop an MLP using PyTorch and evaluate its performance on the FashionMNIST dataset using standard classification metrics. The study further analyzes model performance through ROC curves and loss plots to validate its generalization capability.

B. Objectives

This lab aims to design and evaluate a Multilayer Perceptron (MLP) using PyTorch to classify FashionMNIST images and analyze performance using standard metrics and visualization techniques.

II. Mathematical Background

A. Activation Functions

ReLU (Rectified Linear Unit) is used as the activation function. Defined as $f(x) = \max(0, x)$, ReLU introduces non-linearity while avoiding vanishing gradient problems. The graph of the ReLU function is presented in Figure 1.

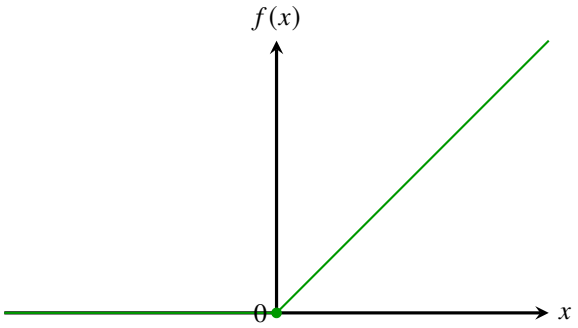


Fig. 1. ReLU Activation Function: $f(x) = \max(0, x)$

B. Encoding

Each grayscale image of size 28×28 is flattened into a 784-dimensional vector. Labels are encoded as integers from 0 to 9 as shown in Table I.

TABLE I
FashionMNIST Label Encoding

Label	Class Name	Label	Class Name
0	T-shirt/top	1	Trouser
2	Pullover	3	Dress
4	Coat	5	Sandal
6	Shirt	7	Sneaker
8	Bag	9	Ankle boot

C. Loss Functions

Cross-Entropy Loss is used:

$$\mathcal{L}(y, \hat{y}) = -\sum_i y_i \log(\hat{y}_i) \tag{1}$$

This function measures the performance of classification models with softmax outputs.

D. Optimizers

Adam optimizer was used due to its ability to adapt learning rates during training, which enhances convergence.

E. Training and Evaluation Loops

Model training includes forward pass, loss calculation, backpropagation, and weight update. Validation loss is computed without weight updates. Evaluation is done on the test dataset.

F. Dataset

FashionMNIST is a labeled dataset of 60,000 training and 10,000 test grayscale images across 10 classes of fashion items. It is publicly available via torchvision.

III. Experiment & Model Implementation

A. Data Preprocessing

The image pixel values were normalized from their original range of $[0, 255]$ to $[-1, 1]$. The training and validation split was 90% for training, 10% for validation.

B. Model Architecture

The MLP consisted of:

- Flatten layer (784 input)
- Dense layer (128 units, ReLU)
- Dense layer (64 units, ReLU)
- Output layer (10 units, logits)

C. Training Configuration

- Batch Size: 64 (training), 256 (validation/test)
- Epochs: 10
- Learning Rate: 0.001
- Optimizer: Adam
- Loss: CrossEntropy

D. Evaluation Metrics

- Accuracy
- Precision (macro)
- Recall (macro)
- F1-Score (macro)
- ROC-AUC per class

E. Code Snippet

```
class MLP(nn.Module):
    def __init__(self):
        super(MLP, self).__init__()
        self.model = nn.Sequential(
            nn.Flatten(),
            nn.Linear(784, 128),
            nn.ReLU(),
            nn.Linear(128, 64),
            nn.ReLU(),
            nn.Linear(64, 10)
        )
    def forward(self, x):
        return self.model(x)
```

Listing 1. Model Definition

IV. Result

The model achieved the following performance on the test dataset:

TABLE II
Performance Metrics

Metric	Value
Accuracy	0.8829
Precision	0.8821
Recall	0.8829
F1-Score	0.8822

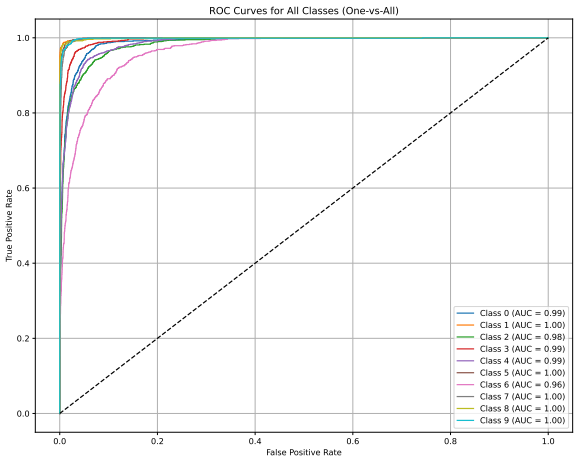


Fig. 2. ROC Curves for All Classes

A. ROC Curves

The ROC-AUC was computed for each class using a one-vs-rest approach. The plotted curves indicate high separability between classes.

B. Training vs Validation Loss

Loss curves indicate smooth convergence without signs of overfitting.



Fig. 3. Training vs Validation Loss

V. Conclusion

This experiment validates that a simple Multilayer Perceptron can classify FashionMNIST images with high accuracy and generalization capability. The ROC and loss plots further support the model's robustness.

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